The MySQL JOIN Clause

The JOIN clause is used to combine rows from two or more tables, based on a related column between them.

Here are the different types of JOINs in MySQL:

- **INNER JOIN: Returns only rows that have matching values in both tables**
- **LEFT JOIN: Returns all rows from the left table, and only the matched rows from the right table**
- **RIGHT JOIN: Returns all rows from the right table, and only the matched rows from the left table**
- **CROSS JOIN: Returns the Cartesian product of two or more tables**

1. INNER JOIN

Definition

Returns only the rows that have matching values in both tables.

Example Tables

Customers

CustomerID	CustomerName
1	Kamalesh
2	Rahul
3	Priya

Orders

OrderID	CustomerID	Product
101	1	Laptop
102	2	Mobile
103	4	Watch

Notice:

- **CustomerID 4 is not present in Customers.**
- **CustomerID 3 has no order.**

Query:

```
SELECT Customers.CustomerName, Orders.Product
FROM Customers
INNER JOIN Orders
ON Customers.CustomerID = Orders.CustomerID;
```

2. LEFT JOIN

Definition

Returns

- All rows from the **left table**
- Matching rows from the **right table**
- If no match exists, NULL is returned.

Query

```
SELECT Customers.CustomerName, Orders.Product
FROM Customers
LEFT JOIN Orders
ON Customers.CustomerID = Orders.CustomerID;
```

Output

CustomerName	Product
Kamalesh	Laptop
Rahul	Mobile
Priya	NULL

Explanation

Priya has no order, but because she is in the **left table**, she still appears.

3. RIGHT JOIN

Definition

Returns

- All rows from the **right table**
- Matching rows from the **left table**
- If no match exists, NULL is returned.

Query

```
SELECT Customers.CustomerName, Orders.Product
FROM Customers
RIGHT JOIN Orders
ON Customers.CustomerID = Orders.CustomerID;
```

Output

CustomerName	Product
Kamalesh	Laptop
Rahul	Mobile
NULL	Watch

Explanation

Order 103 belongs to CustomerID 4, which doesn't exist in Customers.

4. CROSS JOIN

Definition

Returns the **Cartesian Product**.

Every row in the first table is combined with every row in the second table.

Example

Students

Student
A
B

Subjects

Subject
Math
Science

Query

```
SELECT Students.Student, Subjects.Subject
FROM Students
CROSS JOIN Subjects;
```

Output

Student	Subject
A	Math
A	Science
B	Math
B	Science

Explanation

Every student is paired with every subject.

Students = 2 rows

Subjects = 2 rows

Total rows

2 × 2 = 4

What is SELF JOIN?

A **SELF JOIN** means **joining a table with itself**.

👉 Imagine you have **only one table**, but you want to compare one row with another row in the **same table**.

That's why we use **SELF JOIN**.

Example

Employees Table

EmployeeID	Name	ManagerID
1	Alice	NULL
2	Bob	1
3	Charlie	1
4	David	2

Here:

- Alice is the manager.
- Bob's manager is Alice.
- Charlie's manager is Alice.
- David's manager is Bob.

Notice that **employees and managers are stored in the same table**.

```
SELECT e.Name AS Employee,
       m.Name AS Manager
FROM Employees e
LEFT JOIN Employees m
ON e.ManagerID = m.EmployeeID;
```

Output

Employee	Manager
Alice	NULL
Bob	Alice
Charlie	Alice
David	Bob